DKA & HHS — Hyperglycemic Crises

Fluids (balanced) → potassium gate → insulin → close the anion gap, not just the glucose · add dextrose at ~200–250 · find the trigger · overlap-bridge before stopping the drip

1 · Diagnose & distinguish (2024 ADA)

	DKA	HHS
Glucose	≥200 (can be lower — euglycemic DKA)	>600
Ketones/ acidosis	Present; pH/bicarb low (bicarb cutoff 15)	Minimal ketosis; pH >7.3
Osmolality	Variable	>320 (effective)
Mental status	Often alert	Often obtunded

The 2024 ADA consensus lowered the DKA bicarbonate cutoff to 15 and added ketone cutoffs for HHS. Watch for **euglycemic DKA** (SGLT2 inhibitors, pregnancy, starvation) — gap acidosis + ketones with a near-normal glucose.

2 · Fluids & the potassium gate

- **Fluids first: balanced crystalloid** (LR/Plasma-Lyte) now preferred over saline — faster resolution, less hyperchloremic acidosis, shorter stay (2024 consensus).
- Once perfused, switch to half-normal/balanced with potassium and add **dextrose at glucose ~200–250**.
- **Potassium gate before insulin:** if K < 3.3, **hold insulin and replace K first** — insulin drives K intracellularly and can cause fatal hypokalemia.
- K 3.3–5.2: give insulin AND add K to fluids. K >5.2: insulin, recheck, hold K.

Total-body potassium is depleted even when the serum K looks normal/high (acidosis shifts it out). Anticipate the fall once insulin starts.

3 · Insulin & closing the gap

IV insulin infusion for moderate/severe DKA and HHS; **subcutaneous** rapid-acting protocols are an option for **mild, uncomplicated** DKA.

Principle	Detail
Target the gap	Treat to anion-gap/ketone resolution, not glucose alone
Add dextrose	At glucose ~200–250 to keep insulin running
Bicarbonate	Only if pH < 7.0 — not routine
Phosphate	Only if <1.0 mmol/L with weakness/cardiac/resp compromise

DKA resolution = glucose <200 AND two of: bicarb ≥15–18, venous pH >7.3, anion gap normalized. Stopping the drip when only the glucose is normal restarts the ketoacidosis.

Find the trigger (the I's)

- **I**nfection, **I**nfarction (MI), **I**nsulin omission/nonadherence, **I**schemia, **I**atrogenic (steroids, SGLT2i), new-onset diabetes, pregnancy.
- Work-up: glucose, BMP/anion gap, VBG, serum/urine ketones (beta-hydroxybutyrate preferred), osmolality, ECG/troponin, infection screen, HbA1c.

Transition & Do/Avoid

- **DO** overlap SC long-acting insulin with the drip by **1–2 h before stopping** it (prevents rebound ketoacidosis)
- **DO** use balanced crystalloid and respect the potassium gate
- **DO** resolve the gap before declaring DKA over
- **AVOID** insulin when K < 3.3 (replace potassium first)
- **AVOID** routine bicarbonate (only pH <7.0) and routine phosphate
- **AVOID** stopping the insulin drip without overlapping subcutaneous basal insulin

Educational summary for IM residents — not medical advice. Built June 2026. Key refs: ADA/EASD/AACE/JBDS/DTS 2024 Consensus on Hyperglycemic Crises in Adults (Diabetes Care 2024); balanced-fluid evidence.